

1, what is c language

c is a high-level programming language created in the early 1970s by Dennis Ritchie at bell labs. It is one of the most influential language in computer science.

key features of c:

- * fast and efficient - great for system level programming
- * low-level access - can interact directly with memory
- * portable - c programs can run on many different computers with little changes.
- * foundation for many languages - c.

influenced c++, java, c#, python, Go, Rust, and many more

what is c used for

- * Game engines
- * compilers and interpreters

2, Application of C programming

1. Operating systems

c is used to develop major parts of:

- * windows
- * linux
- * macos Because it provides low-level memory access and high performance.

2. Embedded systems

c is the most widely used language in:

- * Microcontrollers
- * IoT devices
- * Medical devices

3. Game Development:

C is used to build:

- * Game engine (like early version of Unreal Engine)
- * High-performance graph modules

4. Compilers and Interpreters

Many programming language compilers are written in C, such as,

- * python (cpython implementation)
- * C++

5. System Software

C is used to develop:

- * Device drivers
- * File systems
- * Networking software

3. What is Variable?

A variable in programming is a named storage location in memory that holds a value which can change during the execution of a program.

Simple Definition

A variable is like a container or box where you store data (numbers, characters, etc.) and use it in your program.

Example in C program

```
C  
int age = 20;
```

Here,

* int → data type

* age → variable

* 20 → value stored in the variable

4, what are different data type in c programming.

1, time_t

* Represents calendar time (number of seconds since the unix epoch : Jan 1, 1970).

* usually an integer or long integer.

* used with functions like `time()`, `difftime()`.

Ex

`time_t current_time;`

`current_time = time(NULL);`

2, struct tm

A structure that holds broken down time (human-readable components).

Defined as:

`struct tm {`

`int tm_sec; // seconds (0-59)`

`int tm_min; // minutes (0-59)`

`int tm_hour; // hours (0-23)`

`int tm_mday; // day of month (1-31)`

`(0-11)`

`int tm_year; // years since 1900.`

`int tm_wday; // day of week (0-6)`

used with functions like `localtime()`, `gmtime()`, `mktime()`.

3, struct timespec

* used for high-resolution time (nanosecond precision)

* contains seconds + nanoseconds

c
struct timespec {

time_t tv_sec; // seconds

long tv_nsec; // nanoseconds

};

4, clock_t

* Represents CPU processor time used by the program.

* used with clock().

c.

clock_t start = clock();

Summary table:-

<u>Type</u>	<u>Description</u>	<u>Precision</u>
time_t	Seconds since epoch	Seconds
struct tm	Human-readable date/time	components (year, month)
struct timespec	High-resolution timestamp	Nanoseconds
clock_t	CPU execution time	clock ticks

5, what is format specifier

In C programming, a format specifier is a code used inside functions like printf() and scanf() to tell the compiler what type of data.

Format specifiers always start with %.

Why format specifiers are needed?

Because C does not know the data type you want to printf or read just from the variable name. Format specifier helps C understand.

* Data type.

* Size

* How to display the value.

Common format specifiers in C.

Data type format specifier

Example.

int

%d or %i

printf

float

%f

printf

double

%lf

char

%c

"

"

String

%s

long int

%ld

long long int

%lld

unsigned int

%u

hexadecimal

%x

octal

%o